5ai	Taking moments about PQ, F x QC = W x DQ B(4.0)(0.15) x 0.60 = 3.0×10^{-4} x 0.20 B = 1.667×10^{-4} = 1.67×10^{-4} T
5aii	Magnetic flux density produced by the solenoid = $1.67 \times 10^{-4} - 5.0 \times 10^{-5} = 1.17 \times 10^{-4}$ Since $B = \mu_o n I$ $1.17 \times 10^{-4} = 4\pi \times 10^{-7} \times 6.0 \times 10^{2}\, I$ $I = 0.155$ A
5bi	Using Fleming's LHR, particle X is the negatively-charged particle.
5bii	Since $Bqv = mv^2/r$ Therefore, $r = mv/Bq$ The particle loses energy / speed / momentum so the radius becomes smaller
5biii	Equal initial radii So equal speed
5ci	As the blade moves, the different parts of the blade experiences changing magnetic flux density, resulting in a <u>change in magnetic flux</u>. By Faraday's law of electromagnetic induction, an emf is induced in the blade. This gives rise to (eddy) currents in the blade and hence heating of the blade. The thermal energy is derived from the mechanical energy (or energy of oscillations) of the blade, so the amplitude decreases.
5cii	resistance
5ciii	The eddy currents will be reduced, hence the opposing (electromagnetic) force will also be smaller.
5di	$\Phi = NBA \cos\theta$ $= 650 \times 2.5 \times 10^{-3} \times 20 \times 10^{-3} \times 35 \times 10^{-3} \times \cos 30°$ $= 9.85 \times 10^{-4}$ Wb turns
5dii	$\Phi = NBA \cos \omega t$ $E = -\frac{d\emptyset}{dt} = -\frac{d(NBA\cos\omega t)}{dt}$ $= \omega NBA \sin \omega t$ $= 2\pi(20) \times 650 \times 2.5 \times 10^{-3} \times 20 \times 10^{-3} \times 35 \times 10^{-3} \sin 40\pi t$ $= 0.143 \sin 40\pi t$ $= 0.14 \sin 40\pi t$ (shown)
5diii	Peak emf halved to 3.5 units And period doubled to 20 units

6a	Electrons from heated cathode bombarded electrons in hydrogen atoms/molecules